1. Analytical Derivation of Sphere-of-Influence Encounter Points

1.1. Problem Statement

Given two Keplerian elliptical orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI}:

$$\|\boldsymbol{A}\boldsymbol{r}_1(E_1) - \boldsymbol{B}\boldsymbol{r}_2(E_2)\| = r_{\mathrm{SOI}} \tag{1}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

1.2. Notation

- a_i, b_i, e_i: semi-major axis, semi-minor axis, and eccentricity of orbit i (with $b_i = a_i\sqrt{1 - e_i^2}$)
- E_i: eccentric anomaly of orbit i
- $\boldsymbol{A} \in \mathbb{R}^{3\times 2}$, $\boldsymbol{B} \in \mathbb{R}^{3\times 2}$: orbital-to-ecliptic rotation matrices with components A_{jk}, B_{jk}
- r_{SOI}: radius of the planet's sphere of influence

1.3. Position in Orbital Plane

For a Keplerian ellipse, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\boldsymbol{r}_i(E_i) = \begin{pmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{pmatrix} \tag{2}$$

1.4. Distance Constraint

Square both sides of the constraint:

$$(\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2)^\top (\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2) = r_{\mathrm{SOI}}^2 \tag{3}$$

Expand the left-hand side:

$$\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{r}_1 - 2\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{B}\boldsymbol{r}_2 + \boldsymbol{r}_2^\top \boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{r}_2 = r_{\mathrm{SOI}}^2 \tag{4}$$

1.4.1. Orthonormality of $\boldsymbol{A}$ and $\boldsymbol{B}$

Since $\boldsymbol{A}$ and $\boldsymbol{B}$ are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}_{2\times 2}, \quad \boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_{2\times 2} \tag{5}$$

Therefore the self-terms simplify:

$$\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{r}_1 = \boldsymbol{r}_1^\top \boldsymbol{r}_1 = |\boldsymbol{r}_1|^2 =: r_1^2 \tag{6}$$

$$\boldsymbol{r}_2^\top \boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{r}_2 = \boldsymbol{r}_2^\top \boldsymbol{r}_2 = |\boldsymbol{r}_2|^2 =: r_2^2 \tag{7}$$

1.4.2. The coupling matrix $\boldsymbol{C}$

Define the 2×2 matrix:

$$\boldsymbol{C} \coloneqq 2\boldsymbol{A}^\top \boldsymbol{B} \tag{8}$$

This encodes the entire mutual orientation of the two orbital planes. Equation 4 becomes:

$$r_1^2 + r_2^2 - \boldsymbol{r}_1^\top \boldsymbol{C}\boldsymbol{r}_2 = r_{\mathrm{SOI}}^2 \tag{9}$$

Note: $\boldsymbol{C}$ is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 2$.

1.5. Newton's Method

We seek to minimise $f(E_1, E_2) \coloneqq r_1^2 + r_2^2 - \boldsymbol{r}_1^\top \boldsymbol{C}\boldsymbol{r}_2 - r_{\mathrm{SOI}}^2$ using Newton's method. Each iteration solves the 2×2 linear system $\boldsymbol{H}\boldsymbol{\delta} = -\nabla f$ for the step $\boldsymbol{\delta} = (\delta E_1, \delta E_2)^\top$, then updates $E_i \leftarrow E_i + \delta E_i$.

1.5.1. Deriving the gradient and Hessian

The objective is $f = r_1^2 + r_2^2 - \boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 - r_{\mathrm{SOI}}^2$, with r_{SOI}^2 constant. We differentiate each part with respect to E_i.

1.5.1.1. Self-terms r_i^2

Using $r_i = a_i(1 - e_i \cos E_i)$:

$$\frac{\partial r_i^2}{\partial E_i} = 2 a_i^2 e_i \sin E_i (1 - e_i \cos E_i) \tag{10}$$

$$\frac{\partial^2 r_i^2}{\partial E_i^2} = 2 a_i^2 e_i (\cos E_i - e_i + 2 e_i \sin^2 E_i) \tag{11}$$

and $\partial r_i^2 / \partial E_j = 0$ for $i \neq j$.

1.5.1.2. Cross-term $\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2$

The position vector Equation 2 can be factored as $\boldsymbol{r}_i = \operatorname{diag}(a_i, b_i) \boldsymbol{p}_i$, where $\operatorname{diag}(a_i, b_i) := \left(\begin{smallmatrix} a_i & 0 \\ 0 & b_i \end{smallmatrix}\right)$ is a diagonal matrix and $\boldsymbol{p}_i$ is a dimensionless position-direction vector:

$$\boldsymbol{p}_i = \begin{pmatrix} \cos E_i - e_i \\ \sin E_i \end{pmatrix} \tag{12}$$

Substituting $\boldsymbol{r}_i = \operatorname{diag}(a_i, b_i) \boldsymbol{p}_i$ into the cross-term and using $(\boldsymbol{X}\boldsymbol{p})^\top = \boldsymbol{p}^\top \boldsymbol{X}^\top$:

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = (\operatorname{diag}(a_1, b_1) \boldsymbol{p}_1)^\top \boldsymbol{C} \operatorname{diag}(a_2, b_2) \boldsymbol{p}_2 = \boldsymbol{p}_1^\top \operatorname{diag}(a_1, b_1)^\top \boldsymbol{C} \operatorname{diag}(a_2, b_2) \boldsymbol{p}_2 \tag{13}$$

Since diagonal matrices are symmetric ($\operatorname{diag}(x, y)^\top = \operatorname{diag}(x, y)$):

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = \boldsymbol{p}_1^\top \operatorname{diag}(a_1, b_1) \boldsymbol{C} \operatorname{diag}(a_2, b_2) \boldsymbol{p}_2 \tag{14}$$

We have a chain of five matrices. Applying associativity ($\boldsymbol{A}(\boldsymbol{B}\boldsymbol{C}) = (\boldsymbol{A}\boldsymbol{B})\boldsymbol{C}$) twice — first to $(\boldsymbol{p}_1^\top \cdot \operatorname{diag}(a_1, b_1)) \cdot \boldsymbol{C} = \boldsymbol{p}_1^\top \cdot (\operatorname{diag}(a_1, b_1) \cdot \boldsymbol{C})$, then to $(\boldsymbol{p}_1^\top \cdot (\operatorname{diag}(a_1, b_1) \cdot \boldsymbol{C})) \cdot \operatorname{diag}(a_2, b_2) = \boldsymbol{p}_1^\top \cdot ((\operatorname{diag}(a_1, b_1) \cdot \boldsymbol{C}) \cdot \operatorname{diag}(a_2, b_2))$ — lets us group the three constant middle factors into a single matrix. We define the **scaled coupling matrix**:

$$\boldsymbol{M} := \operatorname{diag}(a_1, b_1) \boldsymbol{C} \operatorname{diag}(a_2, b_2) = \begin{pmatrix} C_{11} a_1 a_2 & C_{12} a_1 b_2 \\ C_{21} a_2 b_1 & C_{22} b_1 b_2 \end{pmatrix} \tag{15}$$

giving $\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = \boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2$.

Since $\boldsymbol{M}$ is constant, differentiation acts only on $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$. Define the derivative vectors:

$$\hat{\boldsymbol{w}}_i := \mathrm{d}\frac{\boldsymbol{p}_i}{\mathrm{d}E_i} = \begin{pmatrix} -\sin E_i \\ \cos E_i \end{pmatrix}, \quad \hat{\boldsymbol{u}}_i := \begin{pmatrix} \cos E_i \\ \sin E_i \end{pmatrix} \tag{16}$$

To differentiate $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2$ with respect to E_1, note that only $\boldsymbol{p}_1$ depends on E_1. By associativity, $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 = \boldsymbol{p}_1^\top (\boldsymbol{M} \boldsymbol{p}_2)$, which is a dot product of $\boldsymbol{p}_1$ with the constant vector $\boldsymbol{M} \boldsymbol{p}_2$. The derivative of a dot product with one constant factor is:

$$\frac{\partial (\boldsymbol{p}_1^\top (\boldsymbol{M} \boldsymbol{p}_2))}{\partial E_1} = \hat{\boldsymbol{w}}_1^\top (\boldsymbol{M} \boldsymbol{p}_2) = \hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \boldsymbol{p}_2 \tag{17}$$

Similarly, for E_2 only $\boldsymbol{p}_2$ depends on E_2. By associativity, $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 = (\boldsymbol{p}_1^\top \boldsymbol{M}) \boldsymbol{p}_2$, a dot product of the constant row vector $\boldsymbol{p}_1^\top \boldsymbol{M}$ with $\boldsymbol{p}_2$:

$$\frac{\partial ((\boldsymbol{p}_1^\top \boldsymbol{M}) \boldsymbol{p}_2)}{\partial E_2} = (\boldsymbol{p}_1^\top \boldsymbol{M}) \hat{\boldsymbol{w}}_2 = \boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2 \tag{18}$$

The same reasoning applies to higher derivatives, replacing $\boldsymbol{p}_i$ by successive derivatives $\hat{\boldsymbol{w}}_i, -\hat{\boldsymbol{u}}_i$:

$$\frac{\partial^2 (\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2)}{\partial E_1^2} = -\hat{\boldsymbol{u}}_1^\top \boldsymbol{M} \boldsymbol{p}_2, \quad \frac{\partial^2 (\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2)}{\partial E_2^2} = -\boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{u}}_2 \tag{19}$$

$$\frac{\partial^2 (\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2)}{\partial E_1 \partial E_2} = \hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2 \tag{20}$$

1.5.1.3. Combined expressions

Combining $f = r_1^2 + r_2^2 - \boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 - r_{\text{SOI}}^2$ using Equation 10 with Equation 17, and Equation 11 with Equation 19 and Equation 20:

1.5.2. Gradient

$$\frac{\partial f}{\partial E_1} = 2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \boldsymbol{p}_2 \tag{21}$$

$$\frac{\partial f}{\partial E_2} = 2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2 \tag{22}$$

1.5.3. Hessian

The Hessian $\boldsymbol{H}$ is symmetric ($H_{12} = H_{21}$):

$$H_{11} = 2a_1^2 e_1 (\cos E_1 - e_1 + 2e_1 \sin^2 E_1) + \hat{\boldsymbol{u}}_1^\top \boldsymbol{M} \boldsymbol{p}_2 \tag{23}$$

$$H_{22} = 2a_2^2 e_2 (\cos E_2 - e_2 + 2e_2 \sin^2 E_2) + \boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{u}}_2 \tag{24}$$

$$H_{12} = -\hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2 \tag{25}$$

1.5.4. Solving the Newton step

The 2×2 system is solved directly via Cramer's rule:

$$\Delta = H_{11} H_{22} - H_{12}^2 \tag{26}$$

$$\delta E_1 = \frac{\frac{\partial f}{\partial E_1} H_{22} - \frac{\partial f}{\partial E_2} H_{12}}{-\Delta}, \quad \delta E_2 = \frac{\frac{\partial f}{\partial E_2} H_{11} - \frac{\partial f}{\partial E_1} H_{12}}{-\Delta} \tag{27}$$

All coupling terms are bilinear forms $\boldsymbol{x}^\top \boldsymbol{M} \boldsymbol{y}$ with 2D vectors, each requiring 4 multiplies and 3 adds. The five needed dot products ($\hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \boldsymbol{p}_2$, $\boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2$, $\hat{\boldsymbol{u}}_1^\top \boldsymbol{M} \boldsymbol{p}_2$, $\boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{u}}_2$, $\hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2$) can share the intermediate products $\boldsymbol{M}\boldsymbol{p}_2$, $\boldsymbol{M}\hat{\boldsymbol{w}}_2$, $\boldsymbol{M}\hat{\boldsymbol{u}}_2$ (each a 2D matrix-vector multiply).

1.6. Coarse Encounter Interval

Before running Newton's method, we restrict the search to E_1 values where the spacecraft's heliocentric distance overlaps with the planet's distance range $\pm r_{\text{SOI}}$.

1.6.1. Radial overlap condition

The heliocentric distance of the spacecraft is $r_1 = a_1(1 - e_1 \cos E_1)$. An encounter requires:

$$a_2(1 - e_2) - r_{\text{SOI}} \le r_1 \le a_2(1 + e_2) + r_{\text{SOI}} \tag{28}$$

Substituting $r_1 = a_1(1 - e_1 \cos E_1)$ and solving for $\cos E_1$:

$$\frac{a_1 - a_2(1 + e_2) - r_{\text{SOI}}}{a_1 e_1} \le \cos E_1 \le \frac{a_1 - a_2(1 - e_2) + r_{\text{SOI}}}{a_1 e_1} \tag{29}$$

Clamping both sides to $[-1, 1]$ and applying arccos (which reverses the inequality) gives two symmetric intervals $[E_{\text{lo}}, E_{\text{hi}}]$ and $[-E_{\text{lo}}, -E_{\text{hi}}]$ in E_1. If no valid interval exists (i.e. the clamped range is empty), no encounter is possible at any E_1.

1.6.2. Initial guess via parabolic interpolation

For each of the two symmetric E_1 intervals, we evaluate f at three points: the endpoints and midpoint. Here E_2 is estimated by projecting the spacecraft's 3D position onto the planet's orbital plane and inverting the ellipse parametrisation:

$$E_2 = \operatorname{atan2}\left(q_y / b_2, \quad q_x / a_2 + e_2\right) \tag{30}$$

where $\boldsymbol{q} = \boldsymbol{B}^\top \boldsymbol{A} \boldsymbol{r}_1$ is the projection. This matches the eccentric anomaly that $\boldsymbol{q}$ would have if it lay on the planet's ellipse.

Given three samples (x_0, f_0), (x_1, f_1), (x_2, f_2), the minimiser of the interpolating quadratic is:

$$E_1^* = x_1 - \frac{1}{2}\frac{(x_1 - x_0)^2(f_1 - f_2) - (x_1 - x_2)^2(f_1 - f_0)}{(x_1 - x_0)(f_1 - f_2) - (x_1 - x_2)(f_1 - f_0)} \tag{31}$$

Newton's method is then run from the best initial guess across both branches. If the first branch fails to converge to $f \le r_{\mathrm{SOI}}^2$, the second branch is tried.

1.7. Encounter Intervals

The goal is to find the set of (E_1, E_2) pairs where the distance is at most r_{SOI}:

$$\mathcal{R} := \{(E_1, E_2) : f(E_1, E_2) \le r_{\mathrm{SOI}}^2\} \tag{32}$$

1.7.1. E_2 interval for fixed E_1

For a fixed E_1, all terms depending on E_1 are constants. Define $\boldsymbol{v} := \boldsymbol{M}^\top \boldsymbol{p}_1$, so that $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 = \boldsymbol{v}^\top \boldsymbol{p}_2 = v_1(\cos E_2 - e_2) + v_2 \sin E_2$. Then:

$$f(E_2) = \underbrace{r_1^2 + a_2^2 + v_1 e_2 - r_{\mathrm{SOI}}^2}_{\kappa} - (2a_2^2 e_2 + v_1)\cos E_2 - v_2 \sin E_2 + a_2^2 e_2^2 \cos^2 E_2 \tag{33}$$

Applying $\cos^2 E_2 = \frac{1 + \cos 2E_2}{2}$:

$$f(E_2) = \kappa' - (2a_2^2 e_2 + v_1)\cos E_2 - v_2 \sin E_2 + \frac{a_2^2 e_2^2}{2}\cos 2E_2 \tag{34}$$

where $\kappa' = \kappa + a_2^2 e_2^2 / 2$. This mixes first and second harmonics of E_2.

1.7.1.1. Weierstrass substitution

Substituting $t = \tan(E_2/2)$, so that $\cos E_2 = (1 - t^2)/(1 + t^2)$ and $\sin E_2 = 2t/(1 + t^2)$, and using $\cos 2E_2 = 2\cos^2 E_2 - 1$:

$$\cos 2E_2 = \frac{2(1 - t^2)^2}{(1 + t^2)^2} - 1 = \frac{1 - 6t^2 + t^4}{(1 + t^2)^2} \tag{35}$$

Multiplying Equation 34 through by $(1 + t^2)^2$ yields a quartic polynomial in t:

$$P(t) = (1 + t^2)^2 f(E_2) = 0 \tag{36}$$

The real roots $t_1, \ldots, t_k$ of $P(t) = 0$ (with $k \le 4$) correspond to the E_2 boundary values via $E_2 = 2 \arctan t_i$. The encounter intervals are the segments between consecutive roots where $P(t) \le 0$.

1.7.2. E_1 interval for fixed E_2

By symmetry, fixing E_2 and defining $\boldsymbol{w} := \boldsymbol{M} \boldsymbol{p}_2$ gives the analogous expression:

$$f(E_1) = \underbrace{r_2^2 + a_1^2 + w_1 e_1 - r_{\mathrm{SOI}}^2}_{\lambda} - (2a_1^2 e_1 + w_1)\cos E_1 - w_2 \sin E_1 + a_1^2 e_1^2 \cos^2 E_1 \tag{37}$$

The same Weierstrass substitution $s = \tan(E_1/2)$ yields a quartic in s.

1.7.3. Boundary of $\mathcal{R}$

The boundary $\partial\mathcal{R}$ is the level set $f(E_1, E_2) = r_{\mathrm{SOI}}^2$. It can be traced by:

1. Finding any point on $\partial\mathcal{R}$ using Newton's method (§1.5).
2. Following the implicit curve $f = r_{\mathrm{SOI}}^2$ via the tangent direction. The gradient ∇f is normal to the level set, so the tangent is:

$$\boldsymbol{t} = (-\partial f / \partial E_2, \quad \partial f / \partial E_1) \tag{38}$$

A predictor-corrector scheme (Euler step along $\boldsymbol{t}$, then Newton correction back to $f = r_{\mathrm{SOI}}^2$) traces the full boundary.

1.7.4. Time constraint

Eccentric anomalies are linked to time via Kepler's equation:

$$M_i = E_i - e_i \sin E_i = n_i(t - t_{0,i}) \tag{39}$$

where $n_i = 2\pi / T_i$ is the mean motion and $t_{0,i}$ is the epoch of periapsis passage. At a shared time t, the relationship between E_1 and E_2 is:

$$E_1 - e_1 \sin E_1 - \frac{n_1}{n_2}(E_2 - e_2 \sin E_2) = n_1 \left(t_{0,2} - t_{0,1}\right) + \frac{n_1 - n_2}{n_2} M_{0,2} \tag{40}$$

This is a curve in (E_1, E_2) space. An encounter occurs where this curve intersects the feasible region $\mathcal{R}$.